Web Technologies

.....AND SECURITY

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Chapter 2

HTML 1: Overview

Objectives

1. HTML Defined and its History
2. HTML Syntax
3. Semantic Markup
4. Structure of HTML
5. Quick Tour of HTML
6. HTML Semantic Elements

What is HTML?

- **Hypertext Markup Language:** is a markup language for the web that defines the structure of web pages.
  - the most basic building blocks of every website
    - **HTML** is the skeleton, **CSS** is the skin, and **JavaScript** is the circulatory, digestive, and respiratory systems that brings the structure and the skin to life.
  - the standard markup language for creating Web pages
    - **Hypertext**: text that is organized in order to connect related items
    - **Markup:** a style guide for typesetting anything to be printed in hardcopy or soft copy format
    - **Language:** a language that a computer system understands and uses to interpret commands.

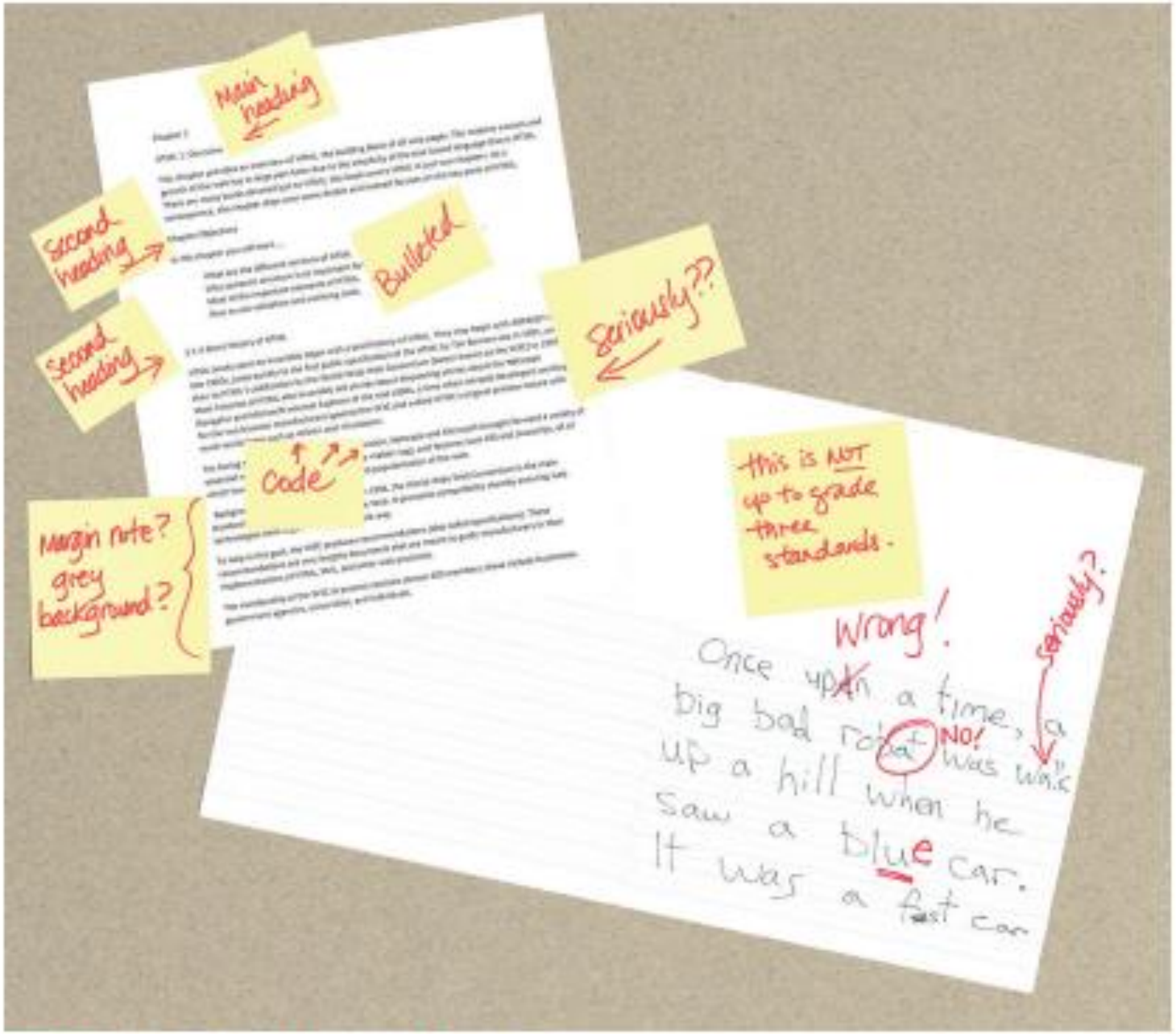


FIGURE 2.1 Sample ad-hoc markup languages

HTML5 Aims

I. Specify unambiguously how browsers should deal with invalid markup.

II. Provide an open, non-proprietary programming framework (via Javascript) for creating rich web applications.

III. Be backwards compatible with the existing web.

- HTML in HTML5 is now a living language: that is, it is a language that evolves and develops over time.
  - previous version : XHTML 1.0, XHTML 2.0, HTML 4.01
  - major browser manufacturers have at least partially embraced HTML5.

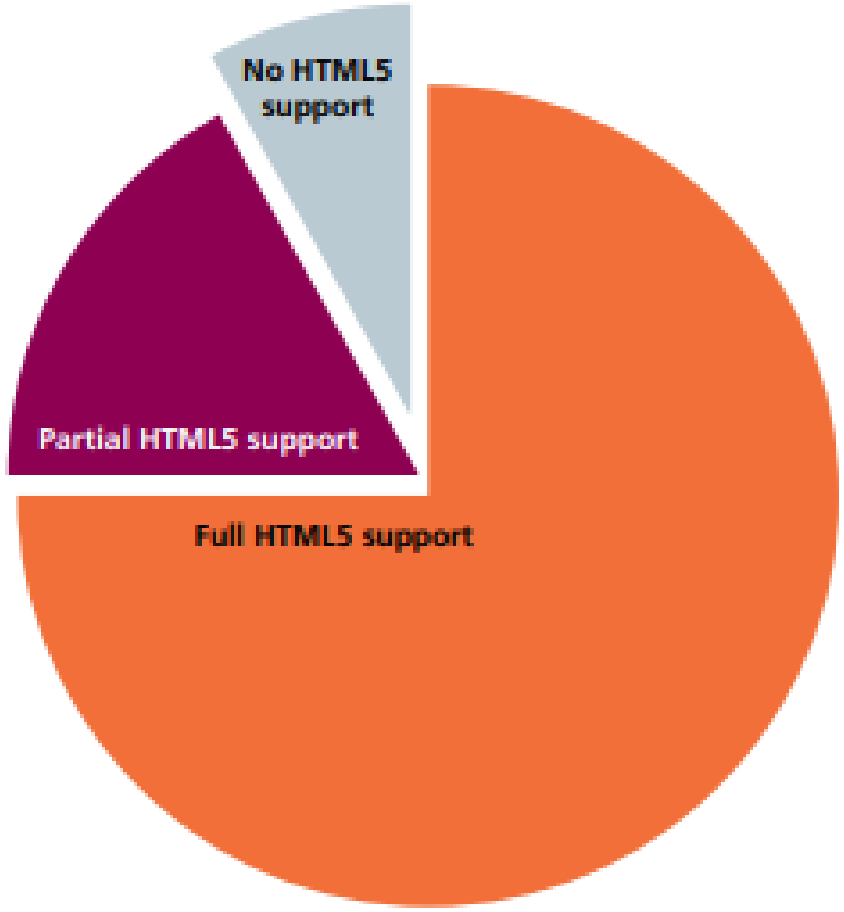


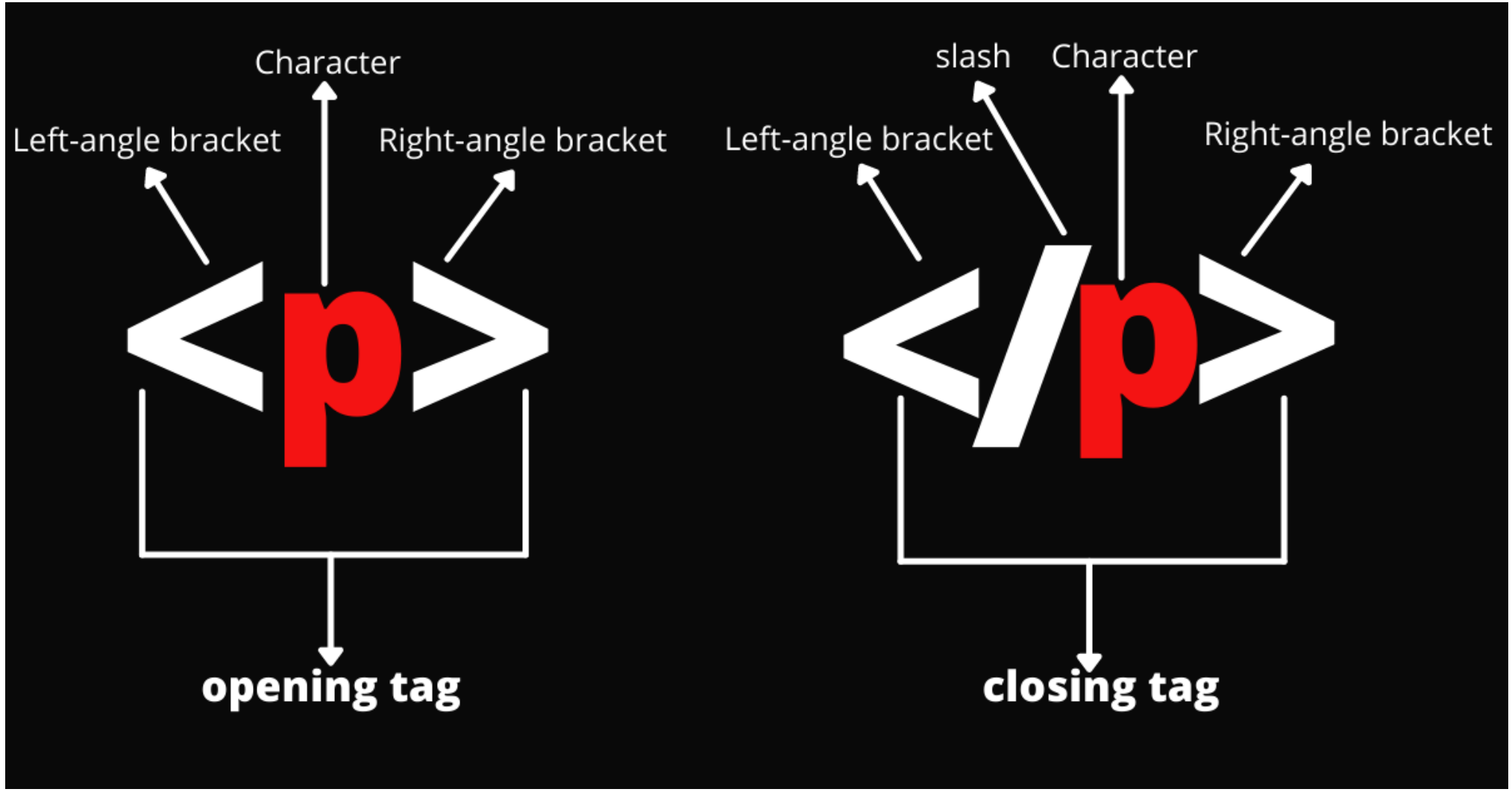
FIGURE 2.3 Browser usage and HTML5 support

HTML Syntax

- HTML consists primarily of ***tags*** that are enclosed inside angle brackets, for example, .
  - **Most tags are paired to indicate the start and end of the text being marked up**;
  - an end tag is formed by including the tag inside the angle brackets with a forward slash, for example, ** **.
- An **HTML tag** is a piece of mark-up language used to indicate the beginning and end of an **HTML element** in an HTML document.
- HTML tags help web browsers convert HTML documents into web pages.
- There are tags to create headings, paragraphs, bolded words, italicized words, and more.
  - the ** ** tag is used to organize text content into paragraph elements and the **<img>** tag is used to embed image elements.

https://www.w3schools.com/tags/default.asp

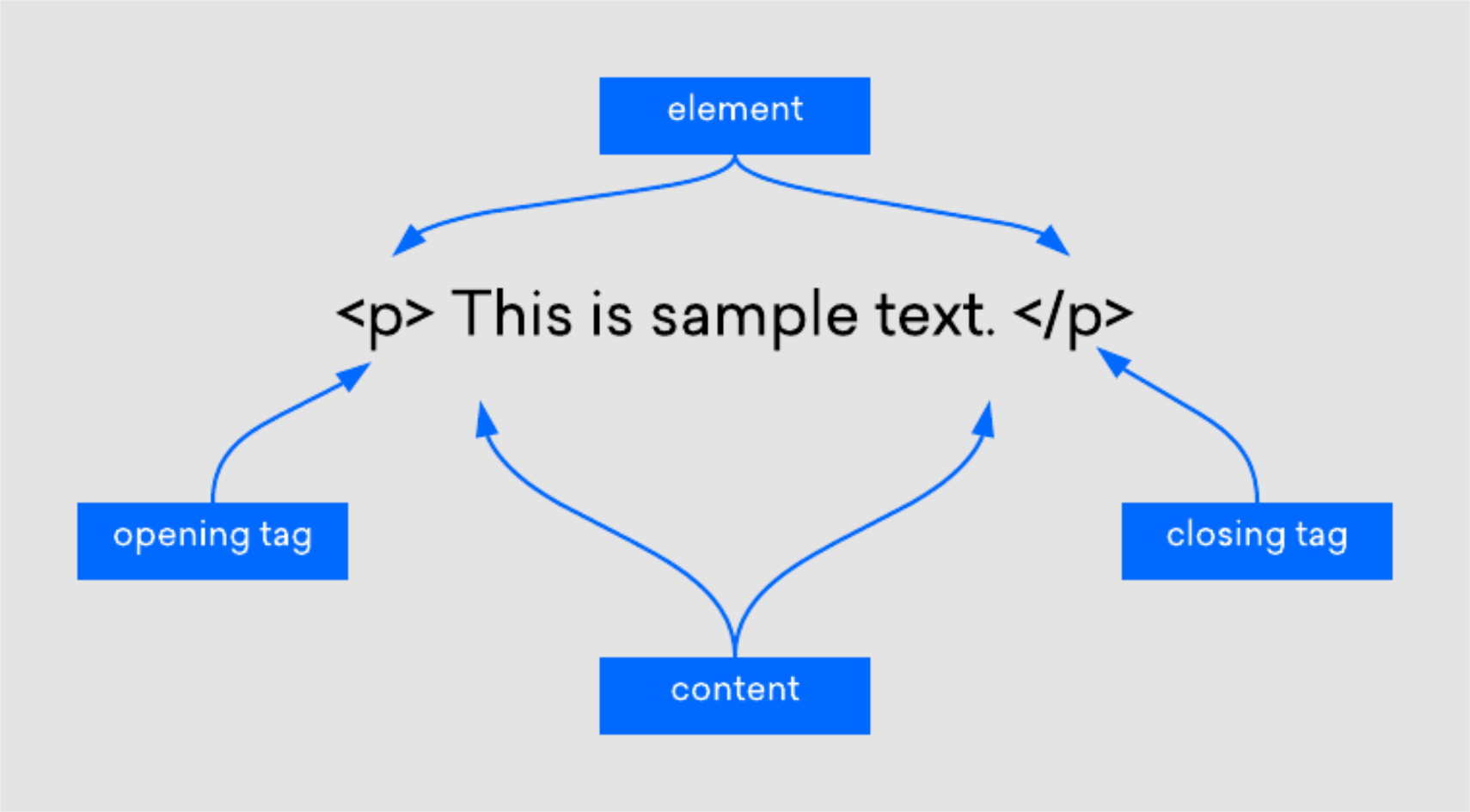
Anatomy of an HTML tag



HTML Elements

- An **element** consists of the opening tag, a character, the content, and a closing tag.
- Some elements are empty –they don't have a closing tag but instead have a source or link to content that you want to embed on the web page.
  - E.g. an empty element is **<img>,** which you use to embed images on a web page.
- HTML elements are often used interchangeably with tags, but there's a small difference between the two:
  - **An element is a combination of the opening and closing tag, and then the content between them.**

Anatomy of an HTML Element

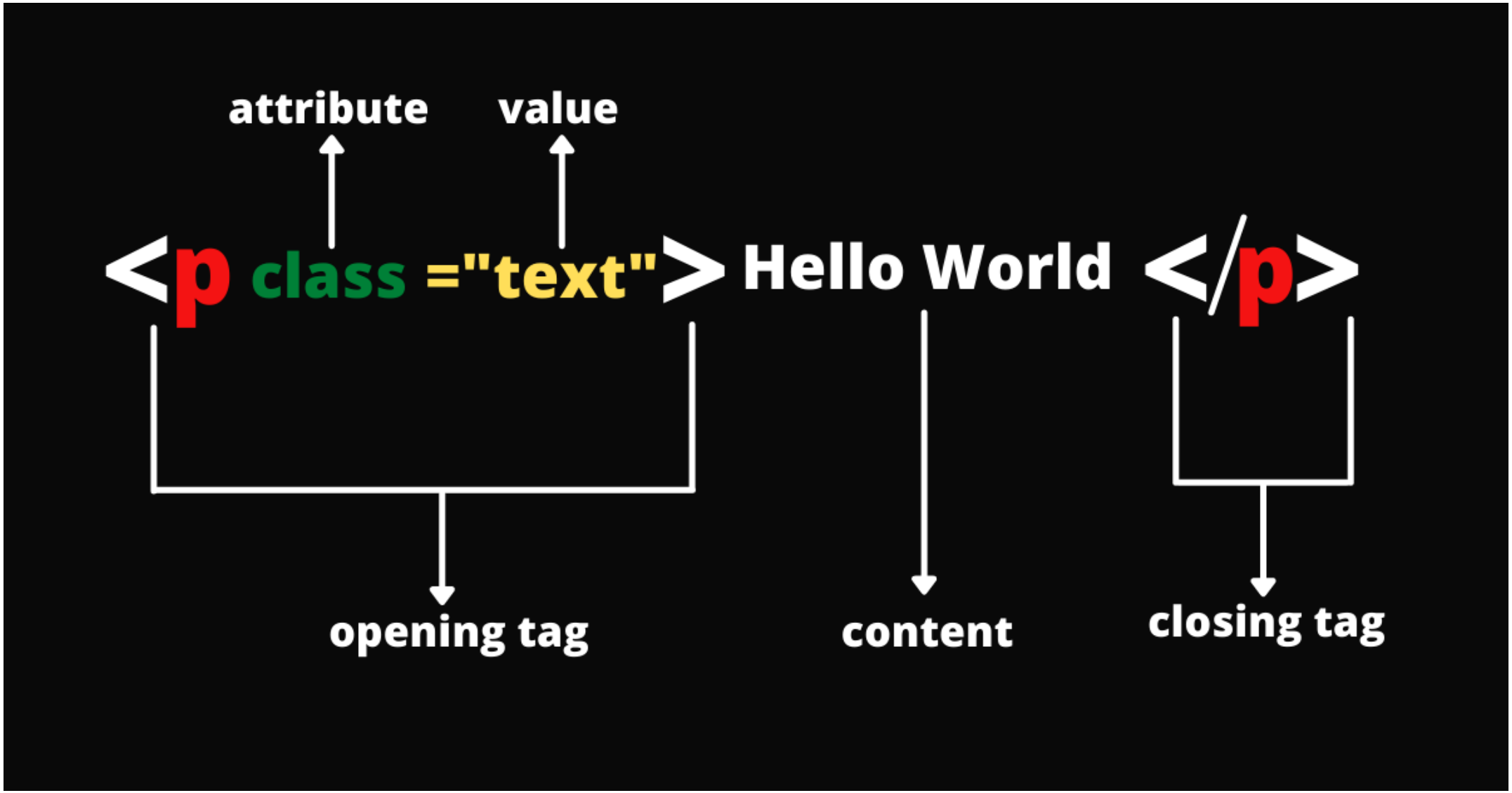


HTML Attributes

- HTML tags also take what are called attributes.
- An **attribute** is used to define the characteristics of an HTML element
  - Attributes are placed inside the element's opening tag.
  - They take values, which convey more information about the element and help you do things such as styling and manipulation with JavaScript.
- The four core attributes that can be used on the majority of HTML elements are –
  - **Id, Title, Class, Style**
- All attributes are made up of two parts – a **name=value** pair that provides more information about the HTML element.
  - The **name** is the property you want to set.
  - The **value** is what you want the value of the property to be set and always put within quotations.

https://www.w3schools.com/TAgs/ref_attributes.asp

Anatomy of an HTML Attributes



Elements and Attributes

Opening Tag

Closing Tag

```
<a href="http://www.centralpark.com">Central Park</a>
```

Element Name

Attribute

Content

May be text or other HTML elements

Trailing Slash

Example empty element

```
<br />
```

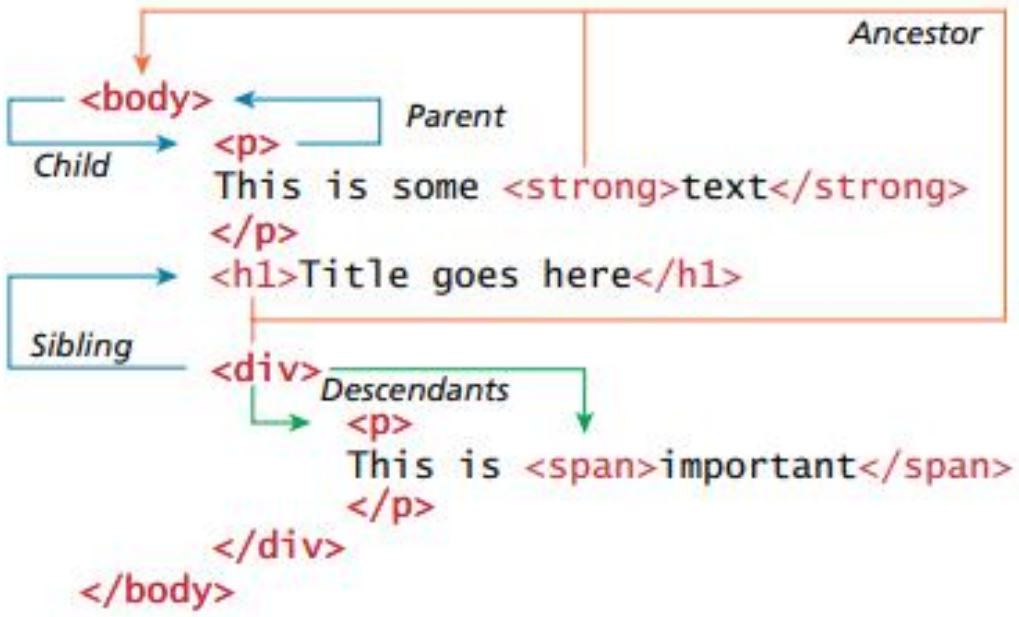
Element Name

Anatomy of a Basic HTML Page

```
<!DOCTYPE html>
<html>

    <head>
        <title>Align Attribute  Example</title>
    </head>

    <body>
        <p align = "left">This is left aligned</p>
        <p align = "center">This is center aligned</p>
        <p align = "right">This is right aligned</p>
    </body>

</html>
```

Nesting HTML elements

- Often an HTML element will contain other HTML elements.
- In such a case, the container element is said to be a parent of the contained, or child, element.
- Any elements contained within the child are said to be **descendants** of the parent element; likewise, any given child element, may have a variety of **ancestors.**



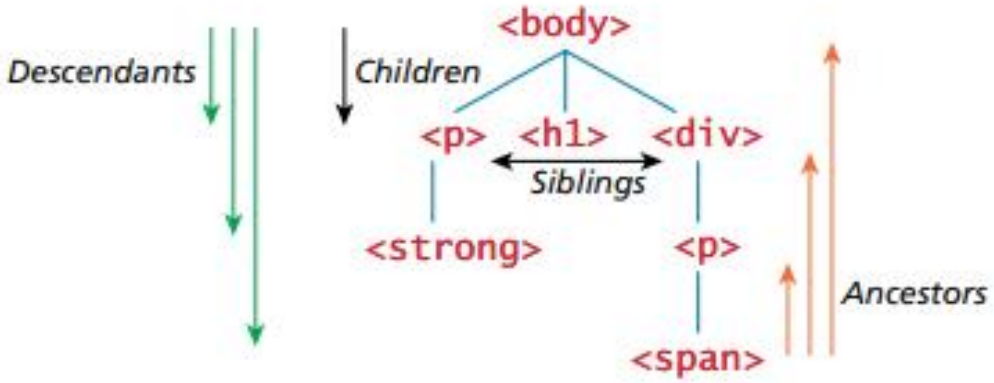


FIGURE 2.5 HTML document outline

Nesting HTML elements

- In order to properly construct a hierarchy of elements, your browser expects each HTML nested element to be properly nested.
  - i.e. a child's ending tag must occur before its parent's ending tag.

Correct nesting

```
<h1>Share Your <strong>Travels</strong></h1>
```

Incorrect nesting

```
<h1>Share Your <strong>Travels</h1></strong>
```

FIGURE 2.6 The proper nesting of HTML elements

Semantic HTML

- **Semantic HTML** means that your HTML tags convey the actual meaning of what they are used for
- With semantic HTML, semantically-neutral tags such as ** ** and **** are frowned upon since semantically more descriptive tags such as **<header>, <nav>, <main>, <section>, <footer>** and **<article>** can do the same thing they do.
- A website that uses semantics becomes more informative, adaptable, and accessible to those who use screen readers to access websites.

Structure Of HTML

- An HTML 4 document is composed of three parts:
  1. a line containing HTML version information,
  2. a declarative header section (delimited by **the HEAD** element),
  3. a **body**, which contains the document's actual content.

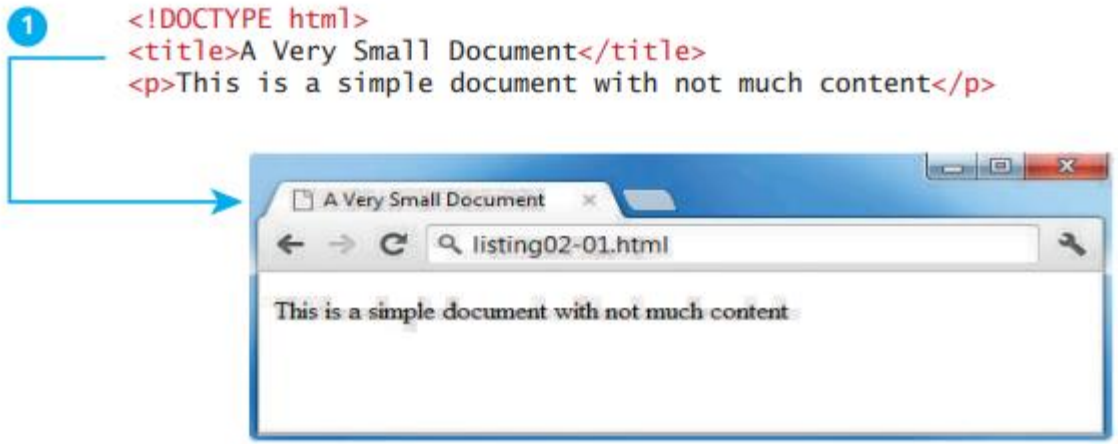


FIGURE 2.8 One of the simplest possible HTML5 documents

```
1  <!DOCTYPE html>
2  <html>
3    <head lang="en">
       <meta charset="utf-8" />                                  5
       <title>Share Your Travels -- New York - Central Park</title>
       <link rel="stylesheet" href="css/main.css" />             6
       <script src="js/html5shiv.js"></script>                   7
     </head>
4    <body>
       <h1>Main heading goes here</h1>
       ...
     </body>
   </html>
```

FIGURE 2.9 Structure elements of an HTML5 document

Structure elements of an HTML5 document

- 1 **DOCTYPE** - short for Document Type - Tells the browser (or any other client software that is reading this HTML document) what type of document it is about to process.
- 2 The **<html>** element is sometimes called the **root element** as it contains all the other HTML elements in the document
- 3 The **<head>** contains descriptive elements about the document
- 4 The **<body>** contains content that will be displayed by the browser.
- 5 the **<meta>** element {in e.g.} declares that the character encoding for the document is UTF-8.
- 6 The example specifies an external CSS style sheet file that is used with this document.
- 7 It also references an external Javascript file

Quick Tour Of HTML

- Why Quick Tour?
  - HTML5 contains many structural and presentation elements—too many to completely cover in this book. Rather than comprehensively cover all these elements, this
  - chapter will provide a quick overview of the most common elements.
  - **Most of the elements and tags will be hands on and you doing the research to meet your practical tasks**

```
<body>
   <h1>Share Your Travels</h1>
   <h2>New York - Central Park</h2>
   <p>Photo by Randy Connolly</p>
   <p>This photo of Conservatory Pond in
      <a href="http://www.centralpark.com/">Central Park</a>
      New York City was taken on October 22, 2015 with a
      <strong>Canon EOS 30D</strong> camera.
   </p>
   <img src="images/central-park.jpg" alt="Central Park" />

   <h3>Reviews</h3>
   <div>
      <p>By Ricardo on <time>September 15, 2015</time></p>
      <p>Easy on the HDR buddy.</p>
   </div>

   <div>
      <p>By Susan on <time>October 1, 2015</time></p>
      <p>I love Central Park.</p>
   </div>

   <p><small>Copyright &copy; 2015 Share Your Travels</small></p>
</body>
```

FIGURE 2.10 Sample HTML5 document

Share Your Travels -- New

Figure02-11.html

Share Your Travels

New York - Central Park

Photo by Randy Connolly

This photo of Conservatory Pond in Central Park New York City was taken on October 22, 2011 with a **Canon EOS 30D** camera.

Reviews

By Ricardo on September 15, 2015

Easy on the HDR buddy.

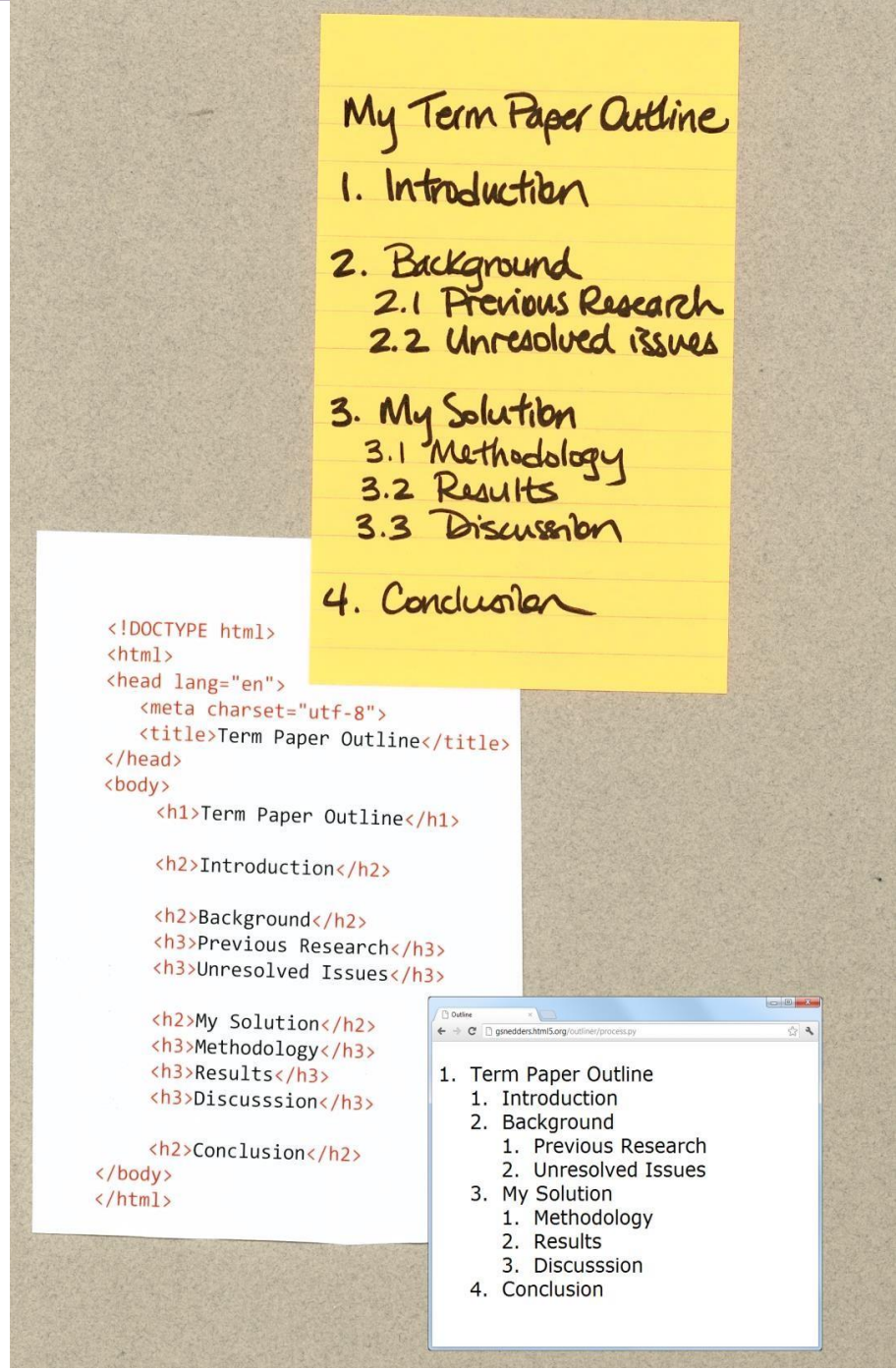
By Susan on October 1, 2015

I love Central Park.

Copyright © 2015 Share Your Travels

Headings

- HTML Provides Six Levels Of Heading (H1, H2, H3, …), With The Higher Heading Number Indicating A Heading Of Less Importance.
  - <h1>, <h2>, <h3>, Etc
- Headings Are An Essential Way For Document Authors Use To Show Their Readers The Structure Of The Document.



Headings

- The browser has its own default styling for each heading level.
- However, these are easily modified and customized via css.
- In practice, specify a heading level that is semantically accurate.
  - Do not choose a heading level because of its default presentation
    - e.g., choosing <h3> because you want your text to be bold and 16pt
  - Rather, choose the heading level because it is appropriate
    - e.g., choosing <h3> because it is a third level heading and not a primary or secondary heading

Paragraphs

- Paragraphs are the most basic unit of text in an HTML document.
- Notice that the **<p>** tag is a container and can contain HTML and **other inline HTML elements**
- **inline HTML elements** refers to HTML elements which only occupy the space bounded by the tags defining the element, instead of breaking the flow of the content.
- An inline element :
  - does not start on a new line.
  - do not cause a paragraph break but are part of the regular "flow" of the text.
  - only takes up as much width as necessary.
  - This is a **** element inside a paragraph.
    - Hello World

Here are the inline elements in HTML:

<a>	<abbr>	<acronym>	<b>	<bdo>	<big>
 	<button>	<cite>	<code>	<dfn>	
<i>	<img>	<input>	<kbd>	<label>	<map>
<object>	<output>	<q>	<samp>	<script>	<select>
<small>			<sub>	<sup>	<textarea>
<time>	<tt>	<var>			

Divisions

- This ** ** tag is:
  - a container element and is used to create a logical grouping of content - which is then styled with CSS or manipulated with JavaScript.
  - defines a division or a section in an HTML document.
  - tag is easily styled by using the class or id attribute.
- Any sort of content can be put inside the ** ** tag!
- It is also **a block element:** A block-level element always starts on a new line, and the browsers automatically add some space (a margin) before and after the element.
- Two commonly used block elements are: **<p>** and ** :**
  - The **<p>** element defines a paragraph in an HTML document.
  - The ** ** element defines a division or a section in an HTML document.
- **Note:** By default, browsers always place a line break before and after the element.

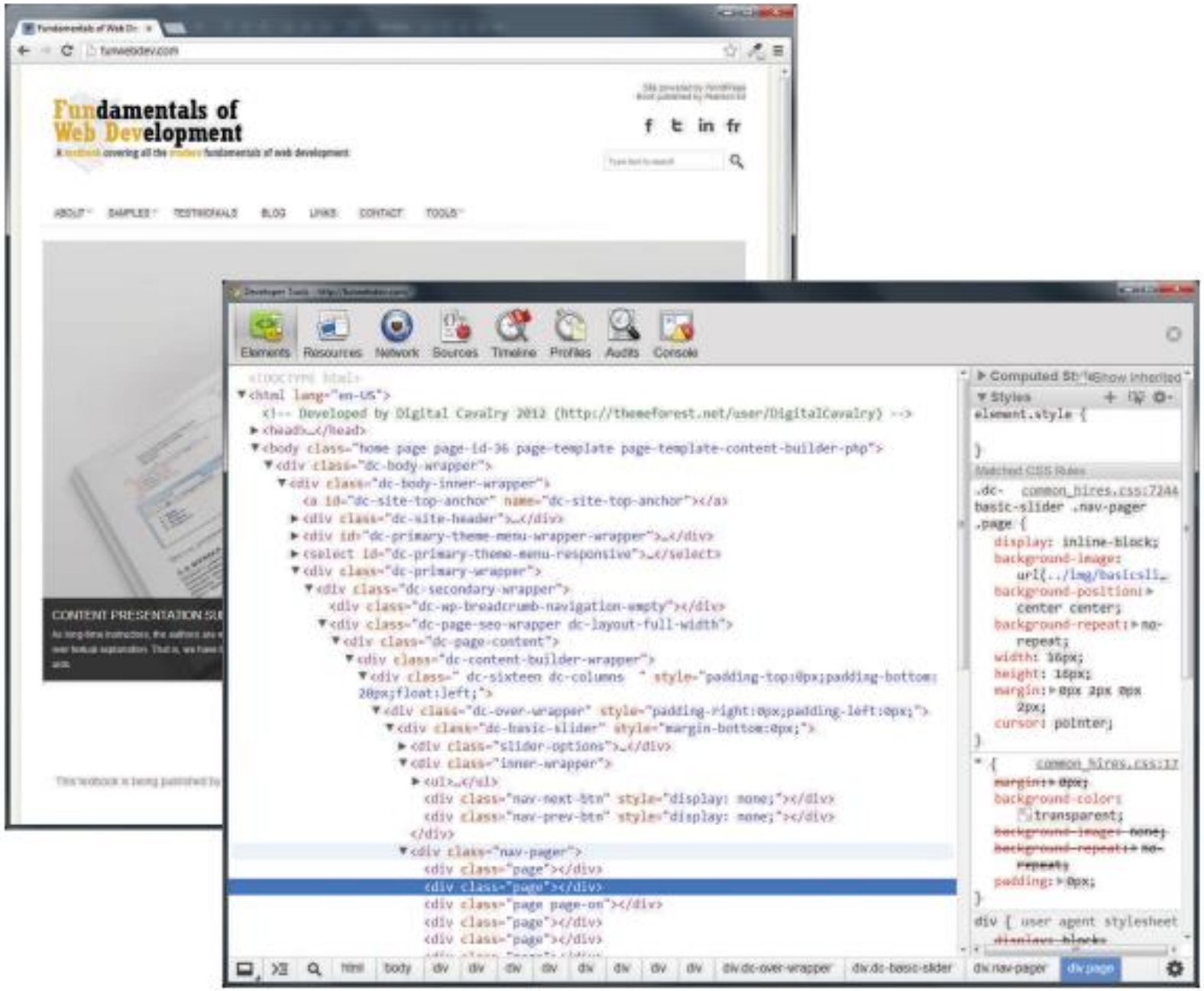


FIGURE 2.14 Using elements to create a complex layout

HTML Links - Syntax <a>

- Links allow users to click their way from page to page
- Hyperlinks are created using the **<a>** element (the "a" stands for anchor).
  - **<a href="url">link text</a>**
  - **The most important attribute of the <a> element is the href attribute, which indicates the link's destination.**
  - The **link text** is the part that will be visible to the reader - Clicking on the link text, will send the reader to the specified URL address.
- A link has two main parts:
  - **the destination**
  - **the label**.

```
<a href="http://www.centralpark.com">Central Park</a>
```

Destination

Label (text)

```
<a href="index.html"><img src="logo.gif" /></a>
```

Label (image)

FIGURE 2.15 Two parts of a link

Types of Links

- You can use the anchor element to create a wide range of links:
  - Links to external sites (or to individual resources such as images or movies on an external site).
  - Links to other pages or resources within the current site.
  - Links to other places within the current page.
  - Links to particular locations on another page.
  - Links that are instructions to the browser to start the user's email program.
  - Links that are instructions to the browser to execute a Javascript function.

Different link destinations

Link to external site

```
<a href="http://www.centralpark.com">Central Park</a>
```

Link to resource on external site

```
<a href="http://www.centralpark.com/logo.gif">Central Park</a>
```

Link to another page on same site as this page

```
<a href="index.html">Home</a>
```

Link to another place on the same page

```
<a href="#top">Go to Top of Document</a>
...
<a name="top">
```

Defines anchor for a link to another place on same page

Link to specific place on another page

```
<a href="productX.html#reviews">Reviews for product X</a>
```

Link to email

```
<a href="mailto:person@somewhere.com">Someone</a>
```

Link to JavaScript function

```
<a href="javascript:OpenAnnoyingPopup();">See This</a>
```

Link to telephone (automatically dials the number when user clicks on it using a smartphone browser)

```
<a href="tel:+18009220579">Call toll free (800) 922-0579</a>
```

FIGURE 2.16 Different link destinations

URL Absolute Referencing

- When referencing a page or resource on an external site, a full absolute reference is required:
  - the protocol (typically, http://),
  - the domain name,
  - any paths, and then finally
  - the file name of the desired resource.
- Here's an example of an absolute URL:
  - https://www.example.com/about/team/

URL Relative Referencing

- URL that only includes the path.
  - The path is everything that comes after the domain, including the directory
- Because relative URLs don't include the entire URL structure, **it is assumed that when linking a relative URL, it uses the same protocol, subdomain and domain as the page it's on.**
- Here's an example of a relative URL:
  - /about/team/

HTML <img> Tag

- The **<img>** tag is used to embed an image in an HTML page.
- Images are not technically inserted into a web page
  - images are linked to web pages.
  - The **<img>** tag creates a holding space for the referenced image.
- The **<img>** tag has two required attributes:
  - The **src** attribute is required, and contains the path to the image you want to embed.
  - The **alt** attribute holds a text description of the image, which isn't mandatory but is incredibly useful for accessibility
- How to insert an image:
  - <img src="img_girl.jpg" alt="Girl in a jacket" width="500" height="600">

HTML Lists

- HTML lists allow web developers to group a set of related items in lists.
- HTML provides three types of lists
  - **Unordered lists:** Collections of items in no particular order; these are by default rendered by the browser as a bulleted list.
    - An unordered list starts with the **<ul>** tag.
    - Each list item starts with the **<li>** tag.
  - **Ordered lists:** Collections of items that have a set order; these are by default rendered by the browser as a numbered list.
    - An ordered list starts with the **<ol>** tag.
    - Each list item starts with the **<li>** tag.

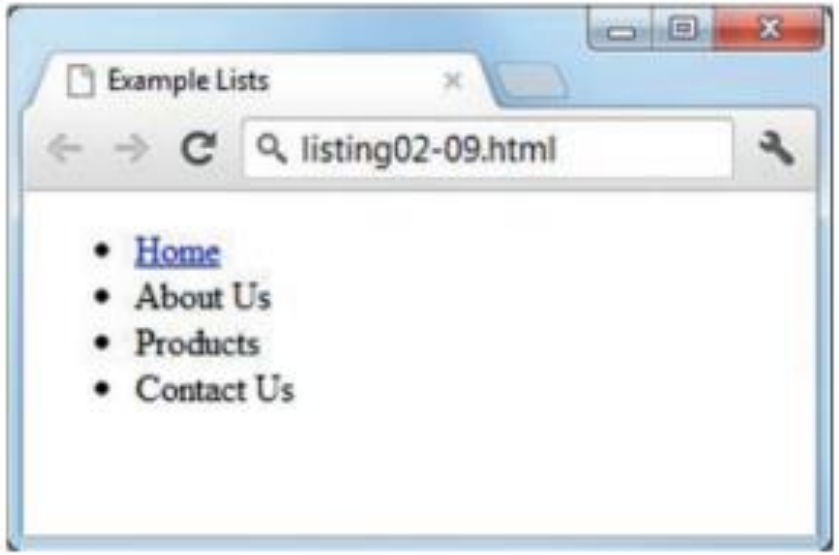
HTML Lists…

- **Definition lists:** Collection of name and definition pairs.
  - This arranges your items in the same way as they are arranged in a dictionary.
  - Definition List makes use of following tags.
  - **<dl>** – Defines the start of the list,
  - **<dt>** – A term
  - **<dd>** – Term definition
  - **</dl>** – Defines the end of the list

Notice that the list item element can contain other HTML elements.

```
<ul>
   <li><a href="index.html">Home</a></li>
   <li>About Us</li>
   <li>Products</li>
   <li>Contact Us</li>
</ul>
```

```
<ol>
   <li>Introduction</li>
   <li>Background</li>
   <li>My Solution</li>
   <li>
      <ol>
         <li>Methodology</li>
         <li>Results</li>
         <li>Discussion</li>
      </ol>
   </li>
   <li>Conclusion</li>
</ol>
```



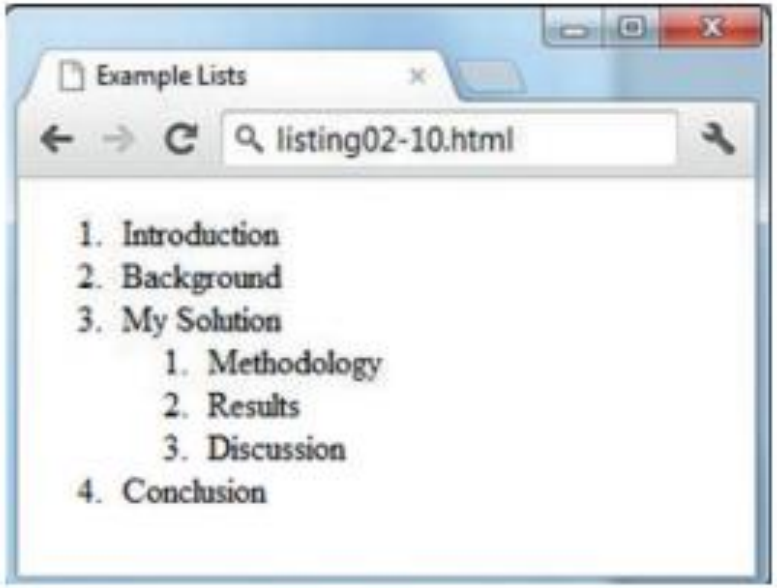


FIGURE 2.19 List elements and their default rendering

```html
<!DOCTYPE html>
<html>

   <head>
      <title>HTML Definition List</title>
   </head>

   <body>
      <dl>
         <dt><b>HTML</b></dt>
         <dd>This stands for Hyper Text Markup Language</dd>
         <dt><b>HTTP</b></dt>
         <dd>This stands for Hyper Text Transfer Protocol</dd>
      </dl>
   </body>

</html>
```

This will produce the following result –

HTML
This stands for Hyper Text Markup Language
HTTP
This stands for Hyper Text Transfer Protocol

HTML SEMANTIC ELEMENTS

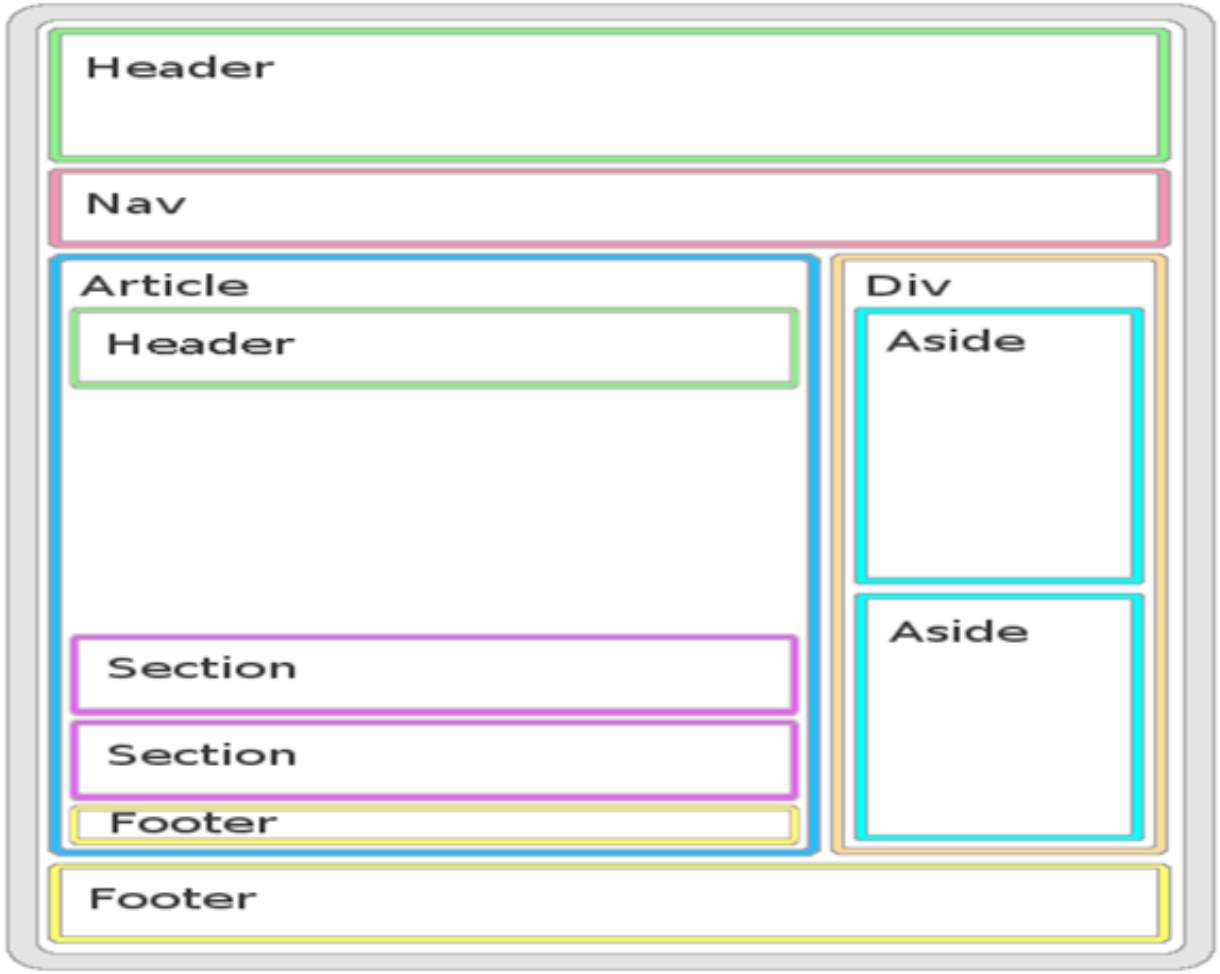
- **Semantic HTML elements** are those that clearly describe their meaning in a human- and machine-readable way.
  - i.e. clearly describes its meaning to both the browser and the developer.
  - Elements such as **<header>, <footer>** and **<article>** are all considered semantic because they accurately describe the purpose of the element and the type of content that is inside them.
- The semantic elements added in HTML5 are:
  - **<article>, <aside>, <details>, <figcaption>, <figure>, <footer>, <header>, <main>, <mark>, <nav>, <section>, <summary>, <time>**
- Examples of **non-semantic elements:**
  - ** ** and **** - These tells nothing about its content.

Semantic vs Non- Semantic HTML Code

```html
<header></header>
<section>
    <article>
        <figure>
            <img>
            <figcaption></figcaption>
        </figure>
    </article>
</section>
<footer></footer>
```

```html
<div id="header"></div>
<div class="section">
    <div class="article">
        <div class="figure">
            <img>
            <div class="figcaption"></div>
        </div>
    </div>
</div>
<div id="footer"></div>
```

HTML5 Template



Why use semantic elements?

- **it is much easier to read:** The easier it is to read and understand that code, the easier it makes your job.
- **It has greater accessibility:** Search engines and assistive technologies (like screen readers) are also able to better understand the context and content of your website
  - meaning a better experience for your users.
- **semantic elements also lead to more consistent code:** When creating a header using non-semantic elements, different programmers might write this a
  - ** , , ,** or simply ** .**
  - By creating a standard semantic element, it makes it easier for everyone.

```html
<article>
  <header>
    <h1>A heading here</h1>
    <p>Posted by John Doe</p>
    <p>Some additional information here</p>
  </header>
  <p>Lorem Ipsum dolor set amet....</p>
</article>
```

Header

Footer

```html
<footer>
  <p>Author: Hege Refsnes</p>
  <p><a href="mailto:hege@example.com">hege@example.com</a></p>
</footer>
```

Navigation

- The <nav> element represents a section of a page that contains links to other pages or to other parts within the same page.

```
<!DOCTYPE html>
<html>
<body>

<header>
<img src="logo.gif" alt="logo" />
<h1>Fundamentals of Web Development</h1>
<nav role="navigation">
<ul>
<li><a href="index.html">Home</a></li>
<li><a href="about.html">About Us</a></li>
<li><a href="browse.html">Browse</a></li>
</ul>
</nav>
</header>

</body>
</html>
```

logo

Fundamentals of Web Development

- Home
- About Us
- Browse

```
<p>This photo was taken on October 22, 2011 with a Canon EOS 30D camera.</p>
<figure>
  <img src="images/central-park.jpg" alt="Central Park" /><br/>
  <figcaption>Conservatory Pond in Central Park</figcaption>
</figure>
<p>
It was a wonderfully beautiful autumn Sunday, with strong sunlight and
expressive clouds. I was very fortunate that my one day in New York was
blessed with such weather!
</p>
```

Figure could be moved to a different location in document ...

But it has to exist in the document (i.e., the figure isn't optional).

FIGURE 2.22 The figure and figcaption elements in the browser